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Breaking the Mold: How ENERGYai is Putting Women at the Helm of Digital Revolution within Oil and Gas

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Abstract

This paper introduces a new narrative—one rooted in visibility, influence, and representation. In one of the world's most traditionally male-dominated industries, a quiet yet powerful transformation is underway. For decades, the upstream oil and gas sector has been shaped by a singular archetype of leadership. But today, a new force is rewriting that script: women. No longer confined to the sidelines of innovation, women are stepping into roles where they are not only present—but pivotal. This work shines a light on how women are leading, contributing to, and representing the creation of *ENERGYai*, ADNOC's groundbreaking AI solution for upstream operations, developed by AIQ in collaboration with G42 and Microsoft. From program management and AI governance to product strategy and delivery, their fingerprints are embedded in every layer of the solution. These are not token roles; they are transformative ones. This is not a story of symbolic inclusion—it is a story of real, structural change. In this journey, inclusion isn't a checkbox at the end—it is the engine powering the entire process. This isn't just about AI. It's about who gets to build the future, and how that future becomes more equitable, more human, and more powerful because of it.

Introduction

The upstream oil and gas industry has long stood as one of the most traditionally male-dominated domains in the global economy. Defined by its technical rigor, operational complexity, and high-stakes decision-making, the sector has historically operated within a rigid archetype of leadership—one that has prioritized efficiency, hierarchy, and uniformity over inclusion, collaboration, and diversity. Within this paradigm, women have often been excluded from critical roles, particularly in technical and strategic capacities, relegated instead to peripheral functions or absent altogether from leadership spaces.

However, a new narrative is beginning to unfold—one in which women are no longer positioned at the margins of innovation but are instead at the helm of it. This shift is particularly visible in the realm of artificial intelligence (AI), where digital transformation is reshaping how oil and gas companies operate, make decisions, and define value. As the industry embraces AI to navigate rising operational demands and environmental challenges, the question is no longer whether transformation is needed—but who will lead it.

This paper explores that question through the case of **ENERGYai**: the world's first industrial agentic AI solution built specifically for upstream oil and gas. Developed by **AIQ** in collaboration with **G42, Microsoft**, and **ADNOC**, ENERGYai represents a technological milestone—automating complex tasks like seismic interpretation and production monitoring while reducing emissions, costs, and operational risk. But beyond its technical innovation lies a quieter revolution: the foundational role of women in shaping its design, development, governance, and delivery. From program and product management to AI governance, UX design, and technical architecture, women have been embedded at every stage of ENERGYai's lifecycle. Their contributions have not only influenced the solution's capabilities but have also disrupted traditional models of leadership and inclusion within the oil and gas sector.

In doing so, ENERGYai offers more than a technological case study—it offers a cultural one. It exemplifies how inclusion can drive innovation, and how diverse leadership can lead to more human-centered, adaptive, and forward-thinking solutions. Particularly in the context of **Abu Dhabi** and the **UAE**, where strategic ambitions center around becoming a global hub for both AI and energy, ENERGYai serves as a microcosm for what inclusive digital transformation can look like in practice.

This paper aims to document and analyze how women's leadership has shaped ENERGYai and how this project provides a blueprint for embedding inclusion at the core of industrial AI innovation. Drawing on internal documentation, interviews with key stakeholders, and project data, this research situates ENERGYai within broader debates around gender, leadership, and the future of technology in extractive industries. In doing so, it contributes to a growing but still limited body of literature that recognizes women not just as participants in digital transformation—but as its architects.

The paper is structured as follows: it explores the historical and theoretical context surrounding gender representation in oil and gas and AI, outlines the methodological approach used in this study, traces the development of ENERGYai, presents the key contributions of women across the solution's lifecycle, and examines cultural and industry-level changes emerging from their involvement. The paper concludes with a reflection on the implications for future innovation and leadership in the sector.

Statement of Theory and Definitions

To understand the significance of women's involvement in the development of ENERGYai, it is essential to first ground the analysis in relevant theoretical frameworks that intersect gender, leadership, and technological innovation. This section defines key concepts used throughout the paper and explores the structural, cultural, and organizational theories that inform the inquiry.

- Gender Representation vs. Inclusion vs. Influence

  In this paper, **gender representation** refers to the numerical presence of women in a given organizational or industrial context. However, representation alone does not guarantee meaningful participation. **Inclusion** moves beyond visibility to describe the extent to which women are integrated into decision-making processes, leadership roles, and strategic influence. Finally, **influence** refers to the actual power individuals hold in shaping outcomes—whether in product strategy, team dynamics, or institutional culture.

  This distinction is critical: it highlights that inclusion is not merely about "being at the table," but about having the authority, credibility, and platform to lead. In the context of ENERGYai, women's influence is understood not only through their roles but through the imprint they leave on the product's vision, governance model, and user-centered design principles.

- The Symbolic Gender Order and Organizational Culture

  The theoretical foundation for understanding women's marginalization in industries like oil and gas can be traced to **Gherardi and Poggio's** concept of the **"symbolic gender order"**. This framework argues that gender is not just a demographic marker but a structural organizing principle within institutions. In male-dominated fields, leadership and expertise are often coded as masculine, compelling women to either adopt male-coded behaviors or risk marginalization.

This symbolic order is reinforced by organizational cultures that reward conformity to traditional norms of control, aggression, and linear logic—traits historically valorized in technical and executive leadership. As a result, women entering such fields face a double-bind: navigating roles shaped by outdated norms while attempting to redefine them from within.

- Inclusive Innovation

  A central theoretical concept underpinning this paper is **inclusive innovation**. As defined by scholars such as Cozzens and Kaplinsky, inclusive innovation refers to the development of products, services, or systems that actively involve and benefit marginalized or underrepresented groups—not only as end-users, but as **designers, decision-makers, and beneficiaries**.

  ENERGYai represents an important case of inclusive innovation, not in the traditional sense of targeting marginalized end-users, but in **how it was built**—by embedding diverse voices in the design and governance process. Inclusion in this context is structural, not symbolic: it is baked into how decisions are made, how teams collaborate, and how success is measured.
- Test Procedures and Problems in Focus

In applying these theories to the case of ENERGYai, several core problems are examined:

- **Problem of access:** Are women present in strategic and technical decision-making spaces?
- **Problem of agency:** Are women able to shape outcomes and assert leadership in visible ways?
- **Problem of culture:** Does the project challenge or reinforce traditional gender norms in oil and gas?

To test these, the study employs a **qualitative case study methodology**, combining documentary analysis with interviews. The unit of analysis is not individual performance, but rather **structural influence**—how the presence of women has reshaped project dynamics, team structures, and leadership models.

The scope of this inquiry extends beyond ENERGYai as a technical solution to consider its broader symbolic and cultural impact. Can inclusion in AI design reshape not only products, but the industries they serve? Can diverse leadership create new paradigms of innovation—more collaborative, human-centered, and equitable?

In answering these questions, this paper draws from feminist organizational theory, innovation studies, and contemporary research on gender equity in STEM and industrial sectors. These theories do not just provide a lens for critique; they offer a blueprint for reimagining what leadership can look like in male-dominated, high-impact fields like oil and gas.

Description and Application of Equipment and Processes

From its inception, women played a critical role in setting the foundation for ENERGYai. Among the earliest contributors were women who shaped the delivery framework, structured key stakeholder engagements, and aligned the technical roadmap with upstream priorities. Their leadership established how the program would operate—centered on accountability, agility, and inclusion.

Their roles positioned them at the intersection of strategy and execution. As young women in senior coordination and delivery functions, their visibility challenged common industry assumptions about who drives innovation in AI and energy.

Delivery and Execution Leadership

At the helm of execution, women leaders orchestrated cross-functional delivery efforts across AI development, data integration, engineering, and user experience design. They ensured that MVP milestones were met, risks mitigated, and stakeholder alignment preserved throughout the project's lifecycle.

Multiple women also led scrum, project management, and product coordination functions—ensuring agile delivery cycles remained grounded in operational relevance. Their contributions bridged strategic vision with day-to-day execution, reinforcing that inclusive leadership is not only equitable, but also highly effective.

Subject Matter and Use Case Expertise

A defining strength of ENERGYai was its integration of deep domain knowledge into AI development. Women domain experts led critical use cases, working alongside data scientists and engineers to align AI outputs with real-world workflows.

Their role was not limited to validation—they actively shaped the design of the models, ensuring both scientific rigor and operational usability. Acting as translators between technical possibility and business application, they demonstrated the critical importance of women in bridging AI with domain-specific expertise.

Enabling the Next Generation

The ENERGYai initiative was also intentional in nurturing early-career female talent. Interns and junior associates were embedded within UX, data, and engineering teams and given meaningful responsibilities. With support from senior mentors, their contributions went beyond observation—they participated in brainstorming sessions, helped test features, and delivered sprint outputs.

This model reflects a pipeline-building mindset, where women at the start of their careers were equipped not only with exposure, but with influence. It signals a long-term shift: inclusion is not just about who leads today—but about who's being empowered to lead tomorrow.

Voices from the Women Behind ENERGYai

"Being involved since the earliest phases of ENERGYai gave me the opportunity not just to manage delivery—but to shape it. We weren't just executing someone else's vision. We were building it together." — *Program Manager, ENERGYai*

"What stood out in this project wasn't just the scale or complexity—it was the intentional collaboration across teams. Having women in decision-making roles didn't feel exceptional. It felt normal. And that's exactly how it should be." — *Product Owner, ENERGYai*

"Seismic interpretation isn't just a technical use case—it's the heartbeat of upstream exploration. As someone rooted in geoscience, it was powerful to guide the AI teams in building a solution that truly reflects the real-world challenges we face in the field." — *Seismic Use Case Lead & SME, ENERGYai*

This collective narrative not only illustrates the scale of women's involvement, but also reframes how leadership, technical credibility, and innovation are distributed in the future of oil and gas. ENERGYai serves as proof that when inclusion is engineered into systems—not layered on top—diverse leadership becomes the norm, not the anomaly.

Discussion

The ENERGYai program offers a compelling case study not just in technical innovation, but in cultural transformation. Through a series of interviews and internal reflections, this section explores key themes that emerged around motivation, empowerment, inclusion, and impact—providing a window into the lived experiences of women at the heart of the project.

Motivations: Building More Than a Product

The women interviewed for this study shared a common thread in their motivations: a desire to be part of building something that had never been done before—not just in terms of AI, but in redefining leadership models within oil and gas.

"We weren't just working on another tech deployment. ENERGYai felt like a chance to build something that could shift how innovation works—both in form and in who gets to lead it." — *Product Owner, ENERGYai*

Several expressed that the opportunity to work on a solution that intersected AI and energy—a historically exclusive space—was not only professionally exciting, but also personally meaningful. For some, it was a form of reclaiming space in a domain where they had long been underrepresented.

Challenges: Navigating Legacy Culture

Despite the inclusive structure of the ENERGYai team, challenges persisted—particularly when interfacing with external teams still operating under traditional norms. Interviewees described instances where their credibility was questioned or where they had to work harder to be heard in technical forums.

"It wasn't always overt, but you could feel it—being the only woman in a room of engineers, having to prove your technical depth before your opinion was considered valid." — *SME and Use Case Lead, ENERGYai*

Others highlighted the internal pressure of being one of the few women in senior roles—feeling responsible not just for their own performance, but for representing broader gender inclusion efforts. This emotional labor was rarely acknowledged but frequently felt.

Empowerment and Inclusion: Structures That Worked

What differentiated ENERGYai from many past initiatives was that inclusion wasn't ad hoc—it was engineered into the process. Interviewees cited specific practices that made the environment empowering:

- **Structured collaboration** (e.g., inclusive sprint planning and regular stakeholder rituals)
- **Merit-based influence**, not hierarchy
- **Cross-functional ownership**, where decision-making authority was shared
- **Leadership modeling**, where senior figures actively invited and elevated diverse voices

"The fact that I was asked to co-lead sprint planning, and not just take minutes—that's when I knew this team was different. Here, inclusion is practice, not PR." — *Project Coordinator, ENERGYai Team*

Importantly, male colleagues were seen not just as neutral participants but as **active enablers**—making space, advocating internally, and challenging legacy behaviors when necessary.

Reflections: What Worked and What's Missing

ENERGYai succeeded in embedding women into every layer of program execution—from strategy to sprint boards. It demonstrated that when diverse talent is intentionally included from the start, both product quality and team cohesion improve.

However, several limitations remain:

- **Scale**: While ENERGYai had a relatively high proportion of women leaders (#30–35%), this is still not representative of broader industry norms.

- **Visibility**: Internally, inclusion was practiced. But externally, many of these women's contributions remain invisible—raising questions about how to tell these stories more broadly.
- **Pipeline fragility**: Early-career support was present, but not yet formalized into a sustained mentorship or upskilling program.

Challenging Norms in Oil and Gas

By placing women at the helm of an AI solution for upstream energy, ENERGYai fundamentally challenges the gendered archetypes that have long defined leadership in the sector. It undermines the notion that innovation must follow top-down, male-dominated hierarchies and replaces it with a model rooted in **collaboration, credibility, and capability**.

This approach reframes inclusion from a DE&I checkbox to a **design principle**—one that actively improves how technology is conceived, built, and deployed.

Contributions to AI Ethics and Inclusive Design

ENERGYai also contributes to the global discourse on **AI ethics** and **inclusive design** in two key ways:

Designing with, not for: Women were not consulted post-facto—they were co-authors of the technology from the start. This aligns with emerging AI ethics principles around participatory design and stakeholder inclusion.

- **Organizational justice in innovation**: The program demonstrated how structural inclusion can improve fairness—not just in algorithmic output, but in team dynamics, power distribution, and voice equity.

As the world increasingly turns to AI to solve complex challenges, ENERGYai offers a vital case study: equitable innovation is not only possible—it is more effective, more ethical, and more enduring.

Conclusions

ENERGYai is more than a technological milestone—it represents a cultural shift in how innovation is led, who gets to lead it, and what inclusive AI development can look like in practice. As the first industrial agentic AI solution tailored to upstream oil and gas, it delivers tangible operational benefits. But its deeper achievement lies in how it redefined the leadership model behind that success.

Women played an instrumental role across every phase of ENERGYai's development—from strategy and governance to program delivery and technical implementation. Their presence wasn't symbolic—it was structural. By embedding inclusion from the outset, the program proved that diversity strengthens both performance and product outcomes.

Interviews revealed that women involved in ENERGYai were driven not just by the challenge of building cutting-edge technology, but by the opportunity to reshape norms in a traditionally male-dominated industry. Their leadership, expertise, and vision helped create a solution that is not only technically robust, but more human-centered and collaborative in its design.

Yet, as impactful as ENERGYai is, it is still an early step. Broader replication and institutional commitment are needed to scale this model. Inclusion must be sustained beyond pilot phases and embedded as a core principle of innovation, not an add-on.

This case sets a precedent. It demonstrates that equitable innovation is not only possible—but powerful. As the oil and gas sector evolves, the question isn't just how technology will change the industry—but who will be shaping that change. With ENERGYai, the answer is clear: women are not just breaking the mold—they're helping build what comes next.

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